PASTA worksheet

Stages	Sneaker company
I. Define business and security objectives	Make **2-3 notes** of specific business requirements that will be analyzed. • *Protect customers personal payment info* • *Ensure payments are processed securely to prevent fraud* • *Provide a reliable shopping experience for buyers and sellers*
II. Define the technical scope	List of technologies used by the application: • *Application programming interface (API)* • *Public key infrastructure (PKI)* • *SHA-256* • *SQL* Write **2-3 sentences** (40-60 words) that describe why you choose to prioritize that technology over the others. I choose PKI because it helps protect customers' personally identifiable info through secure communication. This is incredibly important because if customers' sensitive information gets exposed it could give the company a horrible reputation and cause it to lose a lot of money. Protecting customer information also helps maintain trust with customers.
III. Decompose application	Sample data flow diagram
IV. Threat analysis	List **2 types of threats** in the PASTA worksheet that are risks to the information being handled by the application. • *An employee could misuse their access to customers information to impersonate customers or commit fraud* • *An outside attacker could launch a DDoS attack that overwhelms the sneaker app preventing customers from accessing it or completing purchases*
V. Vulnerability analysis	List **2 vulnerabilities** in the PASTA worksheet that could be exploited. The sneaker app may lack rate limiting or correct traffic

	protection making it easier for attackers to overwhelm the servers with request • *Could there be things wrong with the codebase?* • *Could there be weaknesses in the database?* • *Could there be flaws in the network?*
VI. Attack modeling	Sample attack tree diagram
VII. Risk analysis and impact	List **4 security controls** that you've learned about that can reduce risk.1 prepared statements help prevent sql injection 2. Multifactor authentication adds protection against unauthorized account access 3. Rate limiting and ddos mitigation help protect the website from excessive malicious traffic 4. Secure session management help prevent session hijacking through secure cookies https and proper session expiration

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